

Bruno Trentini

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Education

Ph.D. Computer Science

University of Oxford

Generative Models, Geometric Deep Learning

Supervised by Prof. Michael Bronstein

Oct 2024 – Mar 2028*

United Kingdom

M.Sc. Data Science & Machine Learning

University College London – UCL

Deep Learning, Reinforcement Learning, Graphical Models

First Class / Distinction

Sep 19 – Dec 20

United Kingdom

Robotics & Computation (*Incomplete M.Sc.*)

University College London – UCL

Sep 18 – Nov 18

United Kingdom

Pg.Dip. Telematics & Computer Networks

Univ. Tecnológica Federal do Parana – UTFPR

Best Monographs of the Year (Graded 10/10)

UK NARIC Comparability Certificate

Mar 12 – Sep 13

Brazil

B.Sc. Information Systems

Pontificia Universidade Catolica do Parana – PUCPR

BCI/CREPUQ Scholarship Recipient

UK NARIC Comparability Certificate

Merit - Semester abroad at Bishop's University, Sherbrooke, Canada (Jan 11 - May 11)

Merit - Summer Exchange Program at U. de Technologie de Compiègne, France (Jul 09)

Jan 08 – Dec 11

Brazil

Experience

NVIDIA

Applied Research Scientist

– Researching Generative Models for Molecular Dynamics & Design in collaboration with academic institutions and industry partners.

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United Kingdom

May 22 – Present

GSK

Senior Data Scientist

Researched and implemented multi-objective recommender systems. Reduced Advertisement costs by 55% and the project was shortlisted for an industry award.

United Kingdom

Oct 20 – May 22

KIT-AR United Kingdom
Machine Learning Research Internship Jun 20 – Oct 20
Research project on graph representation learning and human activity recognition using Microsoft HoloLens sensors data.

NVIDIA United Kingdom
AI Startups Ecosystem Manager Dec 18 – Dec 19
Managed the UK AI startups ecosystem, guiding developers towards implementation of machine learning workflows with NVIDIA GPUs and SDKs.

HotelBeds United Kingdom
Commercial Analytics Manager Oct 17 – Oct 18
Applied data science and machine learning techniques to reduce churn, segment customers, and improve financial forecasting.

Alliants United Kingdom
Data & Analytics Consultant Oct 14 – Oct 17
General data analysis and forecasting for clients in the retail and travel industries. Designed new database architectures and wrote data migration scripts.

GVT (Acquired by Telefonica) Brazil
Product Manager II Jun 12 – Sep 14
Worked in the Innovation Lab. Prototyped a gesture-recognition application for new set top boxes using the Microsoft Kinect sensors & SDK in C#.

EY Brazil
Trainee (full-time) Jun 11 – Jun 12
IT General Controls and Finance Auditing. I wrote ACL, Shell, and SQL scripts to identify financial risks and system vulnerabilities.

HSBC Brazil
Trainee Software Analyst (full-time) Jun 08 – Dec 10
Developed COBOL, DB2 and REXX applications for managing large databases of global customer data across IBM z/OS Mainframe Systems.

Biosystems Importadora Brazil
Highschool Internship (part-time) Oct 06 – Mar 07
Created a database of Brazilian Life Sciences researchers.

Ideal Tecnologia Brazil
Apprenticeship (part-time, ad-hoc) Oct 04 – Oct 06
Assembling ID cards in the sealing machine and providing administrative support. I was 14 years old.

Research

Improving Inverse Folding for Peptide Design with Direct Preference Optimization (<https://arxiv.org/abs/2410.19471>)
Oct 2024

Fine-tuning ProteinMPNN using Direct Preference Optimization (DPO) with enhancements to generate diverse and structurally consistent peptide sequences, achieving state-of-the-art structural similarity and improved sequence diversity.

Contributors: Ryan Park, Darren J. Hsu, C. Brian Roland, Maria Korshunova, Chen Tessler, Shie Mannor, Olivia Viessmann, **Bruno Trentini**

Kinesis to Speech: Predicting Industrial Activities Through the Mel-Frequency Cepstrum of Sonified Spatial Data [[Abstract](#)]

Jun 20 – Oct 20

Machine learning research in graph representation learning. The project develops a framework for human activity recognition using Microsoft HoloLens sensor data, introducing a new algorithmic approach for next-step prediction, dimensionality reduction, and multi-modal time-series and 3D data inference. Developed as an intern at [KIT-AR](#).

Advisors: Simon Julier, Felix Mannhardt, Manuel Fradinho Oliveira.

An Empirical Study of using BERT for Authorship Attribution [[Paper](#)]

Jan 20 – Mar 20

Comprehensive set of experiments to distill how BERT responds to several linguistic stimuli at sentence level, identifying how the model functions for text classification and a comparison against LSTM, SVM and BoW.

Advisors: Sebatien Riedl, Tim Rocktäschel, Pasquale Minervini. **Co-authors:** Nicolas Benielli Borrajo, Petar Hristov, Simon Lehnerer

Intelligent Transportation Systems Applied in Traffic Control [[Report](#)]

Mar 13 – Sep 13

Researched how various network architectures and protocols, in addition to genetic algorithms, could impact traffic management in a large city. Different approaches included WiMAX, WLAN, VANET, WAVE, Mesh Networks, and shortest-path algorithms. Elaborated a case study proposing a solution to traffic congestion in Curitiba, a city with 2 million inhabitants in southern Brazil. This was one of the best monographs of the year, and it was awarded the maximum score by the evaluation committee after the viva voce presentation..

Advisor: Prof. Dr. Walter Godoy Junior

Workshops

Machine Learning for Life & Material Science: From Theory to Industry Applications

Accepted at ICML 2024, Pending acceptance at ICML 2025. The proposed workshop, ML4LMS aims to highlight translational machine learning research in biology and chemistry for real-world applications in life and materials science. The goal is to bridge theoretical advances with practical applications and connect academic and industry researchers. Website: www.ml4lms.bio.

Contributors: Aviv Regev, Andrea Volkamer, Bruno Trentini, Cecilia Clementi, Charles Harris, Charlotte Deane, Christian Dallago, Ellen D Zhong, Francesca Grisoni, Jinwoo Leem, Kevin K Yang, Marwin Segler, Michael Martin Pieler, Nicholas James Sofroniew, Olivia Viessmann, Peter K Koo, Pranam Chatterjee, Puck Van Gerwen, Rebecca K. Lindsay, Umberto Lupo, Ying Wai Li

AI for Nucleic Acids (AI4NA) Workshop

Accepted at ICLR 2025. This workshop aims to spotlight nucleic acids as the next frontier for AI research. By bringing together experts from machine learning and biology, we will explore how AI can address key challenges in nucleic acids research, such as RNA tertiary structure prediction, understanding nucleic acid interactions, and designing bespoke RNA/DNA molecules with therapeutic potential. Website: <https://ai4na-workshop.github.io/>

Contributors: Ivona Martinović, Lovro Vrček, Chaitanya K. Joshi, Tin Vlašić, Agata M. Kilar, Bruno Trentini, Max Ward, Maria Brbic, Bryan Hooi, Fran Supek, Pietro Lio, Elena Rivas, Mile Sikic

Memberships

Sr. Organizing Committee - AI4NA Workshop at ICLR	Since 2024
Organizing Committee - ML4LMS Workshop at ICML	Since 2024
Steering Committee - Friends of UCL Computer Science	Since 2024
Professional Member - British Computer Society	Inactive

Cont. Education

Computational Biology – University of Oxford	2024
Oxford Machine Learning Summer School	2024
Building Intelligent Recommender Systems – NVIDIA	2021
Fundamentals of Accelerated Data Science with RAPIDS – NVIDIA	2021

Public Speaking

Imperial College London Mar 25
Opening keynote at the Imperial College’s Future Science Network Symposium on Sustainability and Resilience. Overview of AI in Computational Biology, Protein Folding, and Simulations. Case studies in AgTech, Climate Change, and Genomics.

University College London Jan 25
Invited by the Computer Science Society to talk on Generative AI for Digital Biology and cover topics such as AlphaFold, RosettaFold, and open-source frameworks for biomolecular design.

London Southbank University Nov 24
Overview of Digital Biology and key AI workflows in Biopharma. The talk covered Inverse Folding for Protein Design and Genomics sequencing and alignment.

AI in Healthcare, London Institute of Medical Engineering Apr 24
Overview of latest GPU technologies and biomolecular design workflows with ESMFold, MolMIM and DiffDock and segmentation tasks with Vista-3D.

London X-Linker - Generative AI in Chemistry Dec 23
Overview of Generative AI in Chemistry and NVIDIA Technologies. Mixed audience with 200+ first-year students and experienced developers.

5-HT Chemistry & Health - Generative AI in Chemistry Oct 23
Fundamentals of Generative AI in Chemistry, covering Variational Autoencoders and latest LLM research. Attended by 200+ developers.

Tech Bio X Basel - Innovations from Techbio Startups Oct 23
Overview of NVIDIA technologies in biomolecular design and how startups are utilising these innovations in Drug Discovery.

NVIDIA Generative AI Day for Life Sciences Aug 23
Member of the organizing team. Event with 100 Techbio startup founders and senior scientists, NVIDIA leadership, and investors.

NVIDIA Life Sciences Developer Summit Nov 23
Led the organization team. Attracted 2,000+ developers to talks about AI in protein design, genomics and medical imaging with domain experts and startup founders [[Link](#)].

Explainable AI at GSK May 22
Technical talk based on a case study. Understanding SHAP values' limitations as well as opportunities in explainable AI. [[Abstract](#)][[Slides](#)].

Thompson Sampling at GSK and Data Science Festival Feb 22
Optimal resource allocation applied in online advertisement campaigns with a scalable computational agent using Thompson Sampling and Multi-Armed Bandits informed by Bayesian Inference. Project shortlisted for the 2022 DataIQ Awards in the "Breakthrough with Data" category [[Abstract](#)][[Slides](#)].

Bayesian Statistics at GSK Feb 21
Conjugate and non-informative priors, hypothesis testing, and a drug efficiency case. Lectured to 50 members of the data science division, including Principal Data Scientists and Apprentices from GSK [[Slides](#)].

Overview of Artificial Intelligence at Econet Brazil Dec 19
Introductory presentation to 50 people about AI including paradigms, different models, foundational maths and applications.

Introduction to Deep Learning with GPUs Jul 19

Spoke at multiple NVIDIA events, talking to startups technical leaders and investors. Covered topics on Deep Learning, GPUs and High Performance Computing.

Developer Content

BioNeMo Framework Documentation Sep 23 - Present

Contributed to several sections of the BioNeMo Framework Documentation, a platform providing tools and models for bio-related AI research. Authored tutorials such as “Performing Inference with ESM1-nv and Predictive Modeling with RAPIDS”, , and addressed key areas such as Data Loaders, Fundamentals, and Datasets. [\[link\]](#).

ADMET Prediction and Virtual Screening with Receptor.ai Feb 24

Example of a virtual screening workflow and details about how Receptor.ai combined its Drug Discovery platform with NVIDIA BioNeMo for ADMET tasks [\[link\]](#).

Breakthrough in Functional Annotation with HiFi-NN Dec 23

Basecamp Research, a London-based Bio-AI company, used NVIDIA GPUs to train a Hierarchically Fine-tuned Nearest Neighbor method (HiFi-NN), achieving superior performance in SoTA annotation methods. [\[link\]](#).

Neurosymbolic AI & Knowledge Graphs Nov 23

How Prometheus and NVIDIA integrated the RAPIDS accelerator for Apache Spark into their proprietary knowledge graph management system, Vadalog Parallel, to achieve up to 9x speed ups in Compound/Gene/Disease Similarity tasks with 4.7M edges.[\[link\]](#).

Latest Discoveries in the Developer Summit Oct 22

Non-technical article with uses cases and the future of Life Sciences [\[link\]](#).

Why should I always switch doors in the Monty Hall problem?

Post proving that one should always switch doors in the classic “Monty Hall Problem”. Includes bayesian perspective, visualisations and Python code [\[link\]](#).

7 Resources for Those Wanting to Learn Data Science

Sharing my own experience in growing interest and learning data science and machine learning [\[link\]](#)

Languages

English (Fluent), Portuguese (Native), Spanish (Advanced), French (Basic)

Honours

Selected to Steering Committee of the Friends of UCL Computer Science	2024
Finalist (Shortlisted), DataIQ Awards, “Breakthrough with Data”	2022
Silver Award, Outstanding Contribution in the Data Science Team, GSK	2021
Bronze Award, Contributions to the de-merger activity, GSK	2021
First Class / Distinction, UCL Data Science & Machine Learning	2020
Best Business Solution at the DSS Microsoft/Amex Hackathon	2019
Elected Department Representative, UCL Computer Science	2018
Elected Course Representative, UCL Robotics & Computation	2018

Awarded Professional Membership, British Computer Society	2017
BCI (CREPUQ) Scholarship, \$7,000 grant for semester abroad	2010
Selected to the PUCPR-UTC Academic Exchange	2009
Runner-up, 3rd Mobile Games Workshop hosted by PUCPR	2009